- *Personalization*—You can integrate AI-driven personalization to customize the user experience based on user behavior and preferences.
- *Chat histories*—Consider adding a chat history feature. This allows users to view past conversations, providing context and continuity in their interactions. For that, you want to use a database like MySQL or MongoDB so you can retrieve them by ID. This would involve adding a chat list or a button to view past conversations.
- *Accessibility features*—We can ensure the application is accessible by implementing features like text-to-speech, high-contrast mode, and keyboard navigation.
- *Integrate more AI models*—Consider integrating additional AI models to enhance functionality. For example, consider hosting your own `deepseek-r1` model for a more affordable way to use generative AI in certain interactions.

Remember that for each new feature that you add, you need to include integration tests to ensure that the new feature works well with other parts of the application. This means testing how different components interact with each other.

It's helpful to include scoped logging as it allows you to track specific parts of your application's behavior by logging events within a defined scope. Here is a quick example with Pino:

```
const userLogger= logger.child({ scope: 'user-interaction' });
const aiModelLogger = logger.child({ scope: 'ai-model' });
userLogger.info('User submitted a query');
aiModelLogger.info('AI model generated a response');
```

As you add more features, it's useful to include scoped loggers for those features to make them easier to track. This is just a small part of the overall improvements you can make to your application as you develop it over time.

Summary

- AI-powered applications face unique security threats like prompt injection, model manipulation, and API abuse, requiring specialized security measures beyond traditional software practices.
- Securing AI applications involves a multilayered approach, starting with input validation to prevent malicious data from reaching the AI models.
- Establishing a threat model is crucial for identifying potential vulnerabilities in your application, including public endpoints, user input points, and the sensitivity of the data being processed.
- Server validation is essential to ensure data integrity and security, as it cannot be bypassed by malicious users, unlike client validation, which primarily enhances user experience.
- A security middleware layer acts as a central decision point, analyzing incoming requests for potential threats using various techniques, rate limiting, signature matching, token validation, and machine learning models.